Encapsulation of Fungi To Enhance Bioaugmentation Systems for PAH Remediation

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Poster ID: #19922

Background

Polycyclic aromatic hydrocarbon (PAHs) are a class of carcinogenic compounds that are very persistent in the environment.[1-3] Traditional strategies to remediate them are costly and ecologically disruptive.[4]

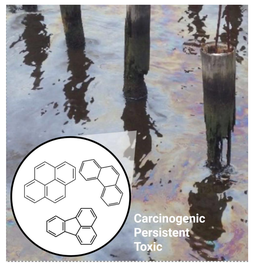


Fig 1. Elizabeth River with visible PAH contamination with example model PAHs

Bioaugmentation is a promising strategy to that **adds** microbes to the contaminated environment to metabolize the PAHs. **However, studies show minimal success under environmentally relevant conditions.** [5]

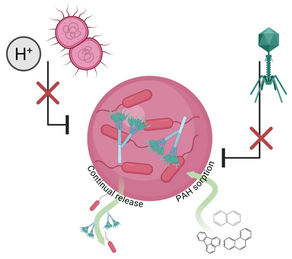


Fig 2. Schematic of the encapsulated microbes showing some of the expected effects

We are encapsulating the microbes in **sodium alginate** to protect them from environmental stressors, promote establishment through continual release, and act as a sink for PAHs. [6]

Project Goals

1. **Assess growth and viability of encapsulated fungal isolates**
2. **Quantify apparent PAH removal (liquid media and the dissolved capsule)**
3. **Measure apparent PAH removal in encapsulated consortia**

Methods

Sodium alginate microbial mixtures are extruded into 0.1 M calcium chloride to produced stable porous capsules. [7]

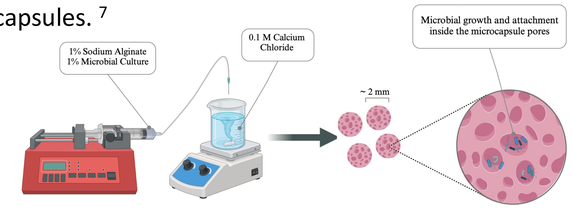


Fig 3. Schematic of the encapsulated microbes showing some of the expected effects

Monocultures
Trichoderma deliquescens
Aureobasidium sp.
Consortia
Trichoderma deliquescens
Novosphingobium aromativans

Cultures grown in R2A media containing 500 ng/mL of each model PAH

Results

Long Term Viability

Encapsulation **promoted viability of Trichoderma** cultures after 8 weeks but had **no effect on Aureobasidium**

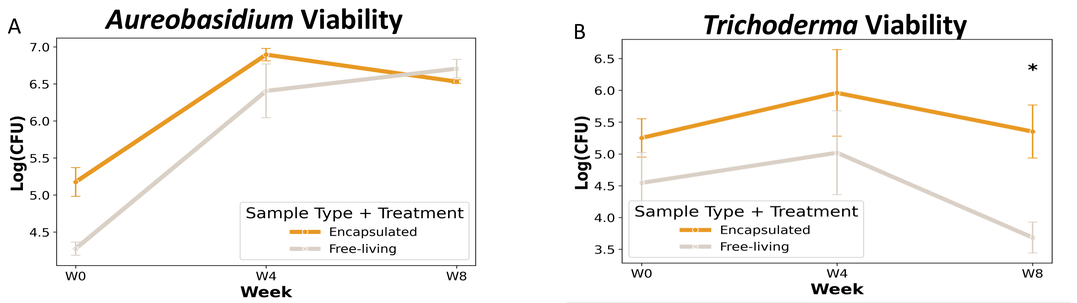
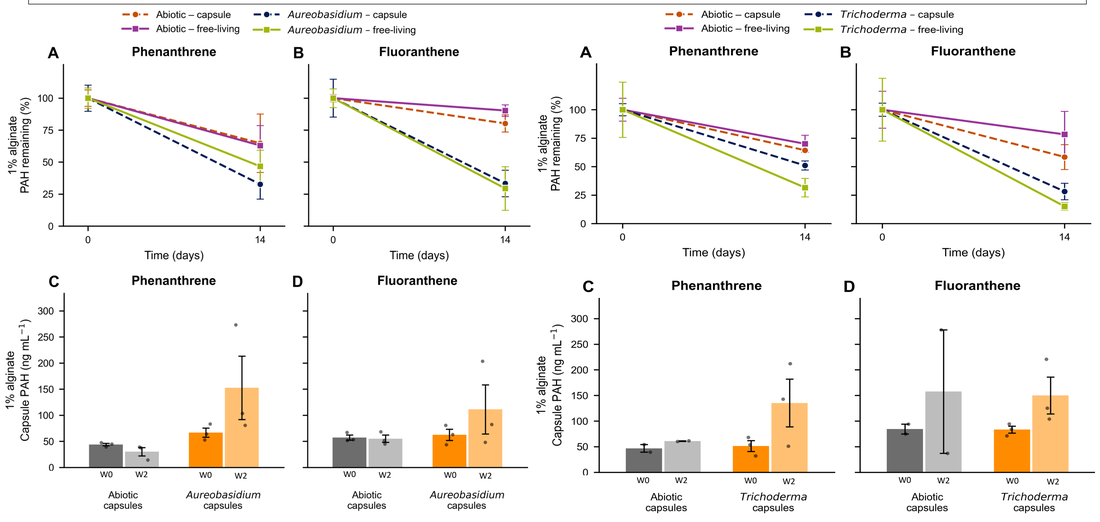


Fig 3 (A-B). Long-term viability of encapsulated and free-living fungal cultures over 8 weeks. Colony-forming units were measured at week 0,4, and 8 for *Aureobasidium* (A) and *Trichoderma* (B). Capsules consisted of 1% alginate extruded at 18 mL h^{-1}. Error bars represent standard error from the mean (n=3).

Apparent removal

PAH **accumulated on capsules** over 2 weeks but showed similar PAH removal trends to free-living cultures.



PAH recovery and capsule-associated retention in **Fig 5 (A-D).** *Aureobasidium* (left) and **Fig 6 (A-D).** *Trichoderma* (right) treatments over 14 days. Columns show phenanthrene and fluoranthene where (A–B) show the percentage of PAH mass remaining relative to day 0 under different conditions and (C-D) (D–F) measure the capsule-associated PAH concentrations recovered from 1% alginate capsules at days 0 and 14.

Consortia effects on capsule sorption

Bacterial capsules **promoted removal of PAHs** whereas fungal capsules **showed a similar trend to the blank** over 8 weeks

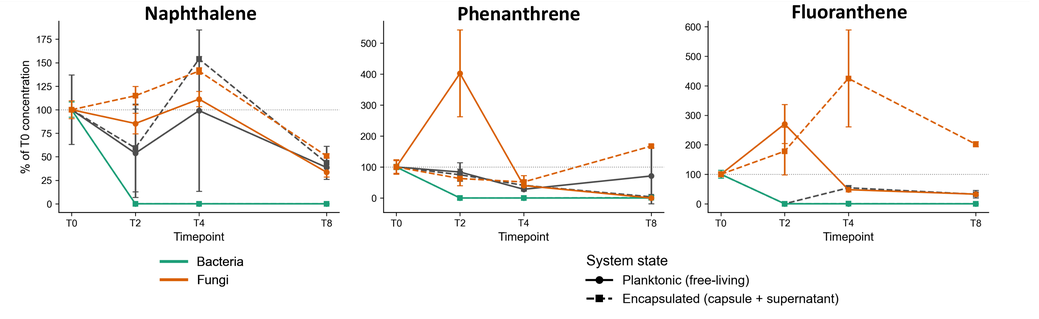


Fig 7 (A-C). Concentrations of PAHs over an 8-week period on the capsules only between fungi (*Trichoderma*), bacteria (*Novosphingobium*), or a consortia.

Conclusions

In lab-controlled conditions

- Encapsulation improves *Trichoderma* survival but had no effect on *Aureobasidium*
- Capsules act as a sink for PAHs
- Pairing fungi with PAH degrading bacteria is key for PAH removal.

Future Directions

- Test capsules in PAH-contaminated sediments under environmentally relevant conditions
- Optimize capsule formulation and consortia to maximize PAH removal
- Trigger fungal PAH metabolism by adding supplements that induce enzyme production

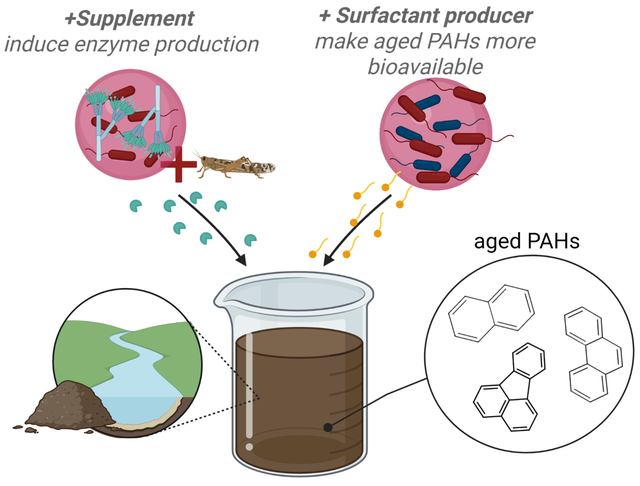


Fig 8 (A-B). Schematic of future experiments incorporating PAH contaminated sediments and capsule formulations hypothesized to help improve removal of aged PAHs

References and Acknowledgments

[1] US EPA (2023), Superfund [2] Di Giulio, R. T., & Clark, B. W. (2015). *J Tox and Env Health* [3] Baek, et al. (1991) *Water Air Soil Pollut*. [4] Seo, J.-S., Keum, Y.-S., & Li, Q. X. (2009). Bacterial Degradation of Aromatic Compounds. [5] Omokhagbor Adams et al., (2020) *International Journal of Environmental Bioremediation & Biodegradation* [6] Babusci & Gunsch. (Under Review). *Journal of Hazardous Materials Advances* [7] Yueng, et al. (2016) *Front. Microbial.*

This material is based upon work supported by the National Institute of Environmental Health Sciences Under Grant No. P42ES010356

Research Portfolio